Pi 5 EM Side-Channel - Phase 1 Findings

Does the Raspberry Pi 5 radiate a data-dependent signal? | 2026-10-17 | R. Boada, adv. S. Sethumadhavan

Goal: show that the Raspberry Pi 5 radiates a measurable, data-dependent RF signal - the first ('memory-bus antenna') milestone of the EM side-channel project - using a handheld spectrum analyzer with the board's memory loop toggled on and off.

Result: YES - but not where we first looked. A single on/off spectrum pair pointed at ~18 MHz, but that was ambient shortwave. With proper averaging, the board's own activity-dependent emission shows up cleanly in UHF at 263.4 MHz and 286.8 MHz (detections at t = 8-10, i.e. 4-5 sigma, +3 to 4.7 dB above the idle floor). The emission is amplitude-modulated by activity, so on/off keying (OOK) at these carriers is the natural scheme going forward - not the low-MHz frequency-shift idea we started with (the memory 'tick' rate barely radiates on this hardware).

Setup: Pi 5 (clocks pinned at 2.4 GHz) running tx_ddr --carrier; tinySA Ultra+ with a whip antenna near the board; no shield yet. Method: interleaved on/off sweeps (carrier toggled each round so ambient cancels), 290 points/sweep, averaged, with a per-bin Welch t-test to separate real emission from ambient.

Challenges faced and how we fixed them

1. Ambient looked like signal. A single on/off pair showed a strong +10 dB line at 18 MHz. It was ambient HF (the 15-20 MHz shortwave/ham band) in an unshielded room. Fix: average many on/off cycles and apply a per-bin significance test; ambient cancels, real emission survives. Caught before any conclusion left the bench.

2. We were looking in the wrong band. The planned low-MHz memory-'tick' tone barely radiates on the Pi 5 (short on-package LPDDR4X). Fix: sweep higher bands with the averaged method, which revealed the true emission at 263.4 / 286.8 MHz (UHF switching/clock activity).

3. Transmitter frequency-parser bug. tx_ddr's --fsk read SI suffixes wrong: '1M' became 1 Hz instead of 1,000,000 Hz. Fix: added an SI-suffix parser (k/M/G) and rebuilt.

4. Antenna swap didn't help. The HackRF ANT500 (75 MHz-1 GHz) performed worse than the stock whip at our frequencies. Fix: kept the whip.

5. Instrument / link friction. The tinySA serial port needed permissions each time it re-enumerated, and the first sweep after a reconnect often returned empty; the Pi transmitter is root-owned and holds the SSH channel open. Fix: warm-up/retry on empty scans, verify carrier state via pgrep before each measurement, and fire-and-forget the remote transmitter.

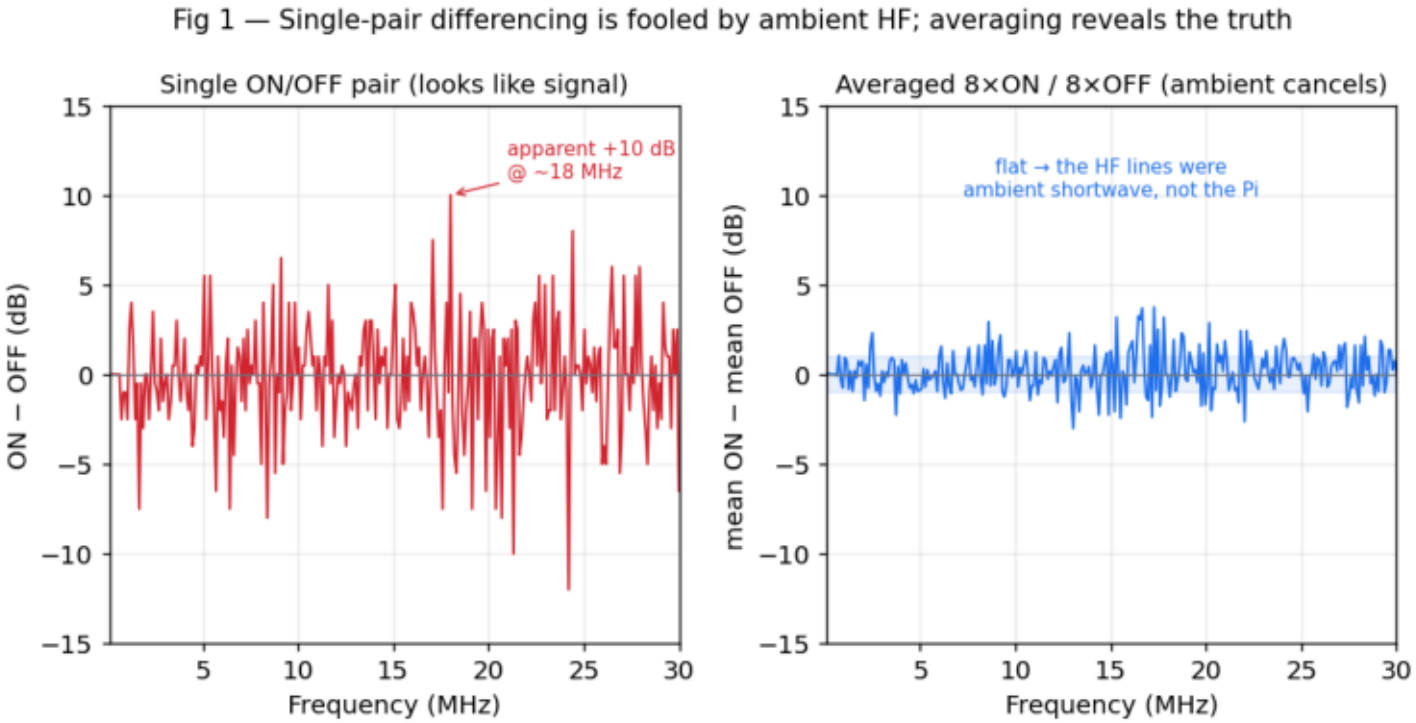


Fig 1 - Why averaging matters. Left: one carrier on/off pair gives a jagged +/-5-10 dB difference with an 'obvious' +10 dB spike at 18 MHz. Right: averaging 8 on + 8 off sweeps collapses the whole band to 0 +/- 1 dB. Ambient signals present in both states cancel; a true board emission would survive. Nothing does, so the 18 MHz line was ambient shortwave, not the Pi.

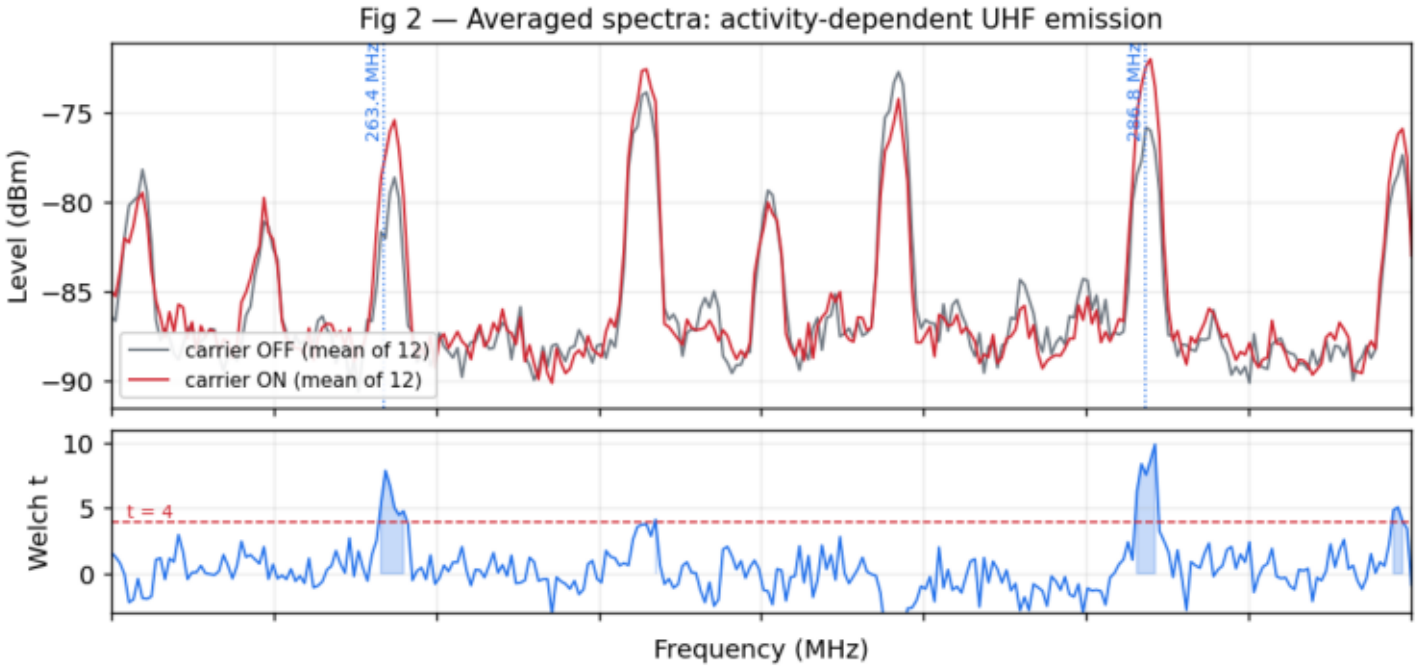


Fig 2 - The real emission, in UHF. Mean of 12 'carrier on' sweeps (red) vs 12 'off' (grey), 255-295 MHz. Several peaks appear in BOTH states (ambient / always-on). Only at 263.4 and 286.8 MHz does 'on' rise above 'off'. Lower panel: per-bin Welch t-statistic; shaded clusters clear t=4 and peak near t=10. Band-wide median |difference| is 0.84 dB, so these are 4-5 sigma detections - a genuine activity-dependent board emission.

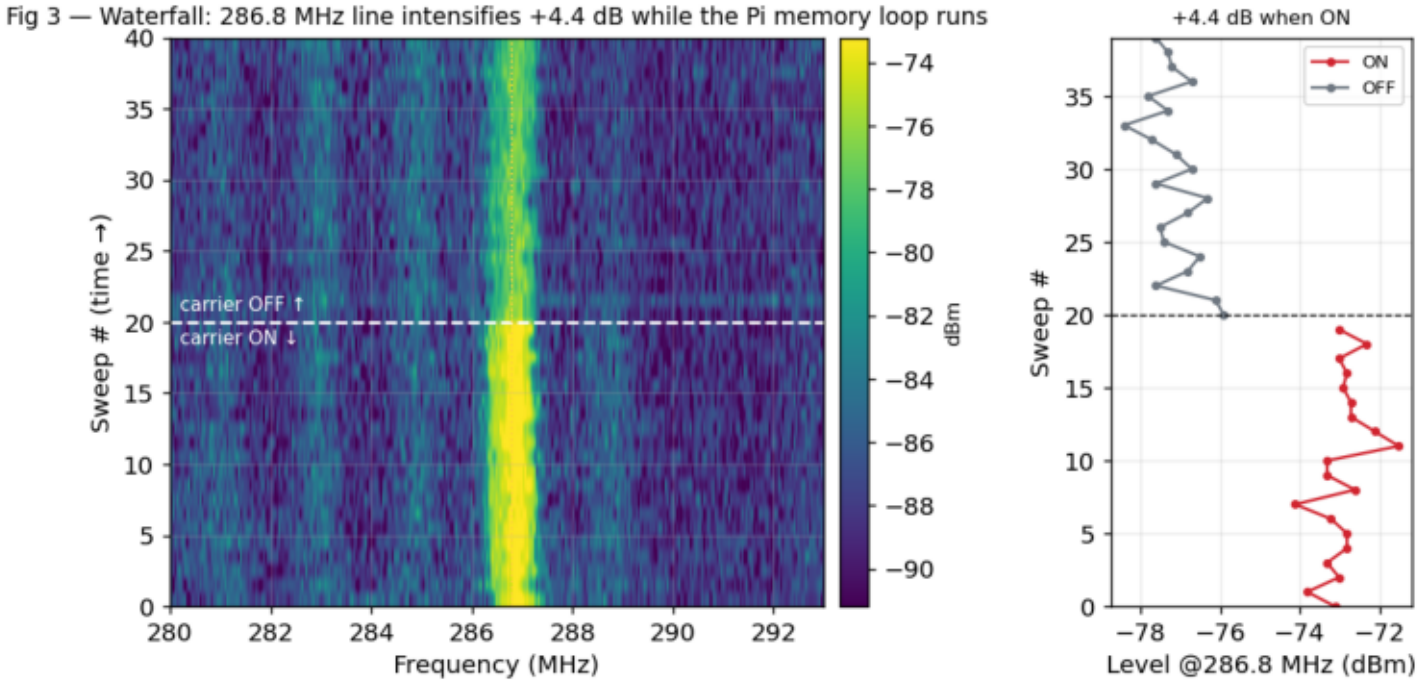


Fig 3 - Watching it switch. Waterfall (frequency x time) around 286.8 MHz, memory loop on for the first 20 sweeps then off for the last 20. The vertical stripe is the emission; it is ~4.4 dB brighter while the loop runs (below the dashed line). The side panel tracks the level at 286.8 MHz per sweep: a clear step at the switch. The line is present at baseline too - likely an always-on clock that activity modulates, which is exactly the hook a side/covert channel would use.